PROJECT REPORT

Build a Model to Predict Stock Market Performance Using LSTM

Project Report on the Results of the Model

MODEL	TARGET	TEST PERIOD	TEST ROWS
LSTM	Close Price	January 2017	20

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Project Focus

Develop an LSTM-based time-series model to learn historical Google stock-price patterns and evaluate predictions on the January 2017 test period.

1. Executive Summary

This project develops a Long Short-Term Memory (LSTM) neural network for predicting Google stock closing prices from historical market data. The objective is to demonstrate a complete, measurable data-science workflow that converts historical observations into sequential learning samples, trains a recurrent neural network, and evaluates its predictions on a separate test period.

The supplied training dataset contains 1,258 observations and the test dataset contains 20 observations for January 2017. The Close price is used as the target variable. The historical Close series is normalized with MinMaxScaler and converted into 60-day rolling sequences. The LSTM model is trained for 50 epochs with a batch size of 32.

MAE	RMSE	MAPE	LOOK-BACK
12.8531	**17.0747**	**≈1.60%**	**60 days**

Key business interpretation

- The model provides a repeatable framework for historical stock-price time-series analysis.
- MAE, RMSE and MAPE quantify the prediction error on the defined test period.
- The Actual vs Predicted chart provides an intuitive view of tracking and error.
- The model is an analytical forecasting exercise and is not a guarantee of future market performance.

2. Business Problem and Objectives

Problem Statement

Stock prices are sequential observations. Previous prices can contain temporal patterns that are not represented when each observation is treated as an independent row. The project therefore uses LSTM, a recurrent architecture designed for sequential data, to learn relationships across historical observations.

Business Challenge

The challenge is to prepare market data without leaking future information, create a consistent historical window, train a model, and measure how closely its predictions match actual prices during an unseen evaluation period.

Project Objectives

1. Load and understand the supplied Google stock datasets.
2. Prepare the Close price as the target variable.
3. Perform exploratory analysis of price and volume behaviour.
4. Normalize the target series using MinMaxScaler.
5. Create 60-day sequences for LSTM training.
6. Build and train a multi-layer LSTM model.
7. Prepare test sequences for January 2017.
8. Generate predicted closing prices.
9. Evaluate predictions using MAE, RMSE and MAPE.
10. Communicate the results through an Actual vs Predicted visualization.

3. Data Understanding and EDA

The dataset contains six fields: Date, Open, High, Low, Close and Volume. The training file contains 1,258 rows and the test file contains 20 rows. January 2017 is used as the test period.

Column	Role	Description
Date	Time index	Trading date
Open	Feature	Opening price
High	Feature	Highest price
Low	Feature	Lowest price
Close	Target	Closing price to be predicted
Volume	Feature	Trading volume

Data Preparation Observations

- The Close field was converted to numeric format after removing comma formatting where needed.
- Close was explicitly defined as the prediction target.
- No future test observations were used to fit the training scaler.
- The chronological order of the time series was preserved.

EDA Performed

- Closing-price trend visualization.
- Trading-volume trend visualization.
- Closing-price distribution histogram.
- Basic minimum, maximum, mean and median statistics.

The EDA provides context on historical movement and variation before the series is transformed into normalized sequences for deep-learning training.

4. Analytical Approach and Data Pipeline

End-to-End Workflow

1. Data Loading
2. Data Understanding
3. Preprocessing and Cleaning
4. Exploratory Data Analysis
5. **Feature Scaling**
6. **Sequence Creation**
7. **LSTM Model Building**
8. Model Training
9. Test Data Preparation
10. **Prediction**
11. **Model Evaluation**
12. **Actual vs Predicted Analysis**

Sequence Design

A look-back window of 60 trading days is used. Each training sample therefore contains the previous 60 scaled Close values and the next Close value is the prediction target. After applying the look-back window, the training set contains 1,198 sequence samples.

Scaling Strategy

MinMaxScaler transforms the training Close values to the 0-to-1 range. The scaler is fitted on training data and then reused for test preparation and inverse transformation of predictions.

Test Preparation

To predict the January 2017 observations, the model receives the required historical context from the end of the training series. This creates 20 test sequences with the shape required by the LSTM.

5. LSTM Model Design and Training

Why LSTM?

LSTM is a recurrent neural-network architecture designed to process sequential information. It can learn temporal patterns from historical observations and use the sequence context to estimate a subsequent value.

Model Architecture

Component	Configuration	Purpose
Input	60 time steps × 1 feature	Historical Close sequence
LSTM 1	50 units	Learn sequence representations
Dropout	20%	Regularization
LSTM 2	50 units	Learn deeper temporal patterns
Dropout	20%	Regularization
LSTM 3	50 units	Final recurrent representation
Dropout	20%	Regularization
Dense	1 unit	Predicted Close value

EPOCHS	BATCH SIZE	OPTIMIZER	LOSS
50	32	Adam	Mean Squared Error

The model is trained on 1,198 sequence samples. The training-loss curve is used to observe the learning behaviour during the 50 training epochs.

6. Model Results and Evaluation

The trained LSTM is evaluated on the 20 January 2017 observations. Predictions are converted back to the original price scale before calculating the metrics.

Metric	Result	Interpretation
MAE	12.8531	Average absolute error in the original price scale
RMSE	17.0747	Error metric that gives more weight to larger errors
MAPE	≈1.60%	Average absolute percentage error on the test period

Result Interpretation

The MAPE of approximately 1.60% summarizes the average percentage difference between actual and predicted closing prices for the evaluated test period. MAE and RMSE express error in the original price scale. RMSE is higher than MAE because larger errors contribute more strongly to the squared-error measure.

Actual vs Predicted

The Actual vs Predicted line chart shows how the model's predictions move relative to the observed January 2017 closing prices. Close overlap indicates smaller error, while wider separation indicates larger error for an observation.

These results describe the supplied test period only. They should not be interpreted as a guarantee of future market accuracy.

7. Business Insights and Use Cases

Insights from the Modelling Exercise

- Historical market observations can be transformed into supervised learning sequences for deep learning.
- A fixed 60-day window gives the model a consistent representation of recent historical context.
- The model output can be communicated through both quantitative metrics and visual comparisons.
- The workflow is reproducible and can be used as a baseline for further experiments.

Potential Business Use Cases

- Forecast values can be displayed alongside historical prices in analytical dashboards.
- Research teams can compare LSTM performance with simpler time-series baselines.
- Model performance can be monitored using MAE, RMSE and MAPE over multiple evaluation periods.
- The pipeline can be extended with additional market features and scheduled retraining.

Decision-Support Boundary

- The model does not guarantee future prices.
- A single test month is not sufficient to establish long-term generalization.
- Historical Close alone does not capture all information affecting financial markets.
- Investment decisions require broader information, risk assessment and appropriate validation.

Recommended Next Steps

11. Compare against simple persistence/last-value baselines.
12. Test 30-day, 60-day and 90-day look-back windows.
13. Use a separate validation period for hyperparameter tuning.
14. Experiment with multivariate OHLCV inputs.
15. Evaluate multiple market periods and regimes.

8. Limitations, Risks and Improvement Roadmap

Limitations

- The current target model focuses on the Close price series.
- The test window contains only 20 observations.
- News, macroeconomic indicators, fundamentals and other external drivers are not explicitly included.
- Neural-network results can vary with model architecture, initialization and hyperparameters.
- Unexpected market events can create behaviour not represented in historical training data.

Improvement Roadmap

Improvement	Potential benefit	How to validate
Hyperparameter tuning	May reduce validation error	Compare validation metrics
Alternative look-back windows	May capture different temporal patterns	30 vs 60 vs 90 days
Multivariate LSTM	Adds OHLCV context	Compare with univariate model
Early stopping	May reduce unnecessary training	Monitor validation loss
Longer test periods	Improves robustness assessment	Evaluate multiple periods
Baseline comparison	Shows incremental value of LSTM	Compare with simple forecasts

Risk-aware Model Use

For a finance application, model error should be monitored together with uncertainty and downside risk. Any operational use should be supported by repeated out-of-sample validation and monitoring across different market regimes.

9. Conclusion

This project successfully implements an end-to-end LSTM-based time-series prediction workflow for Google stock closing prices. The workflow covers data understanding, preprocessing, exploratory analysis, normalization, sequence creation, LSTM model training, test preparation, prediction and quantitative evaluation.

The final model uses three LSTM layers with Dropout regularization and a single Dense output neuron. It learns from 60-day historical sequences and is trained for 50 epochs with a batch size of 32 using the Adam optimizer and mean squared error loss.

TRAINING ROWS	TRAINING SEQUENCES	TEST ROWS	MAPE
1,258	1,198	20	≈1.60%

Final Results

On the supplied January 2017 test period, the model achieved an MAE of 12.8531, an RMSE of 17.0747 and a MAPE of approximately 1.60%. These values quantify the observed prediction error on the defined test period.

Final Conclusion

The project demonstrates how LSTM-based deep learning can be applied to financial time-series analysis through a structured and measurable modelling process. The strongest outcome is the reproducible pipeline: historical data is prepared without fitting the scaler on future test observations, temporal context is preserved through 60-day sequences, and model predictions are evaluated quantitatively and visually.

The results should be viewed as evidence of model behaviour on the supplied test period rather than as a guarantee of future market performance. A production-oriented extension should focus on longer evaluation periods, validation-based hyperparameter tuning, baseline comparisons and richer market features.